

Tutorial-6

1) Without using calculator find value of

- (a) $\sin(\frac{\pi}{12})$ (b) $\tan(75^\circ)$ (c) $\cos(105^\circ)$
(d) $\tan(15^\circ)$ (e) $\sin(75^\circ)$.

2) Prove that $\sin \alpha \cos(\beta - \alpha) + \cos \alpha \sin(\beta - \alpha) = \sin \beta$.

3) Prove that

$$\sin(\frac{\pi}{3} + A) \cos(\frac{\pi}{3} + B) - \cos(\frac{\pi}{3} + A) \sin(\frac{\pi}{3} + B) = \sin(A - B)$$

4) Evaluate without using calculator.

(a) $\frac{\tan(32^\circ) + \tan(88^\circ)}{1 - \tan(32^\circ) \cdot \tan(88^\circ)}$ (b) $\frac{\tan(66^\circ) + \tan(69^\circ)}{1 - \tan(66^\circ) \tan(69^\circ)}$

5) Prove that

$$\sin(A + \frac{\pi}{6}) - \sin(A - \frac{\pi}{6}) = \cos A.$$

6) Prove that

$$\sin(A + B) \cdot \sin(A - B) = \cos^2 B - \cos^2 A.$$

7) Prove that

$$\frac{\sin(A - B)}{\cos A \cdot \cos B} + \frac{\sin(B - C)}{\cos B \cdot \cos C} + \frac{\sin(C - A)}{\cos C \cdot \cos A} = 0$$

8) Prove that

$$\frac{1 - \tan 2\theta \cdot \tan \theta}{1 + \tan 2\theta \tan \theta} = \frac{\cos 3\theta}{\cos \theta}.$$

⑨ Show that

$$\tan 3A - \tan 2A - \tan A = \tan 3A \cdot \tan 2A \cdot \tan A.$$

⑩ Show that

$$\tan\left(\frac{\pi}{4} - \theta\right) = \frac{1 - \tan \theta}{1 + \tan \theta}$$

⑪ If $A + B = \frac{\pi}{4}$ then show that
 $(1 + \tan A)(1 + \tan B) = 2$.

⑫ If $\tan x = \frac{5}{6}$ and $\tan y = \frac{1}{11}$
prove that $x + y = \frac{\pi}{4}$.

⑬ If $\tan(x + y) = \frac{3}{4}$ and $\tan(x - y) = \frac{8}{15}$
prove that $\tan 2x = \frac{77}{36}$, Also find $\tan 2y$.

⑭ If $\tan(x + y) = \frac{1}{2}$ and $\tan(x - y) = \frac{1}{3}$
find (i) $\tan 2x$ (ii) $\tan 2y$.

⑮ If $\sin \alpha = \frac{12}{13}$, $\cos \beta = \frac{3}{5}$, $\frac{\pi}{2} < \alpha < \pi$,
and $\frac{3\pi}{2} < \beta < 2\pi$ find $\cos(\alpha + \beta)$